

HDOC DAY-31

2D ARRAY OPERATIONS USING MENU-DRIVEN C PROGRAM

Name	Dhruv Pandey
SAP ID	590037660
Batch	15
Subject	C Programming
Activity	Matrix Sum, Row Sums, Transpose

Problem Statement

Write a C program to perform the following menu-driven operations on a given 2D array: (1) find the sum of all elements in a matrix, (2) find the sum of each row and store it in an array, (3) find the transpose of the matrix, and (4) exit.

1. Algorithm

1. Start the program.
2. Read the number of rows and columns.
3. Read all matrix elements using nested loops.
4. Display the menu repeatedly until Exit is selected.
5. For choice 1, add every matrix element and display the total sum.
6. For choice 2, calculate each row sum, store it in rowSum[], and display the row-sum array.
7. For choice 3, form the transpose by assigning $\text{transpose}[j][i] = a[i][j]$, then display it.
8. For choice 4, terminate the program.
9. For any other choice, display an invalid-choice message.
10. Stop.

2. Test Case Table

Test	Input / Operation	Expected Result
1	Matrix: 1 2 3 / 4 5 6; Choice 1	Sum = 21
2	Same matrix; Choice 2	Row sums = [6, 15]
3	Same matrix; Choice 3	Transpose: 1 4 / 2 5 / 3 6
4	Choice 4	Program exits

3. C Program

```
#include <stdio.h>

#define MAX 100

int main()
{
    int a[MAX][MAX], transpose[MAX][MAX];
    int rowSum[MAX];
    int rows, cols;
    int i, j, choice, sum;

    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &rows, &cols);

    printf("Enter the matrix elements:\n");
    for (i = 0; i < rows; i++)
        for (j = 0; j < cols; j++)
            scanf("%d", &a[i][j]);

    do
    {
        printf("\n===== MATRIX OPERATIONS =====\n");
        printf("1. Sum of all elements\n");
        printf("2. Sum of each row\n");
        printf("3. Transpose of matrix\n");
        printf("4. Exit\n");
        printf("===== \n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                sum = 0;
                for (i = 0; i < rows; i++)
                    for (j = 0; j < cols; j++)
                        sum = sum + a[i][j];
                printf("Sum of all elements = %d\n", sum);
                break;

            case 2:
                printf("Row sums:\n");
                for (i = 0; i < rows; i++)
                {
                    rowSum[i] = 0;
                    for (j = 0; j < cols; j++)
                        rowSum[i] = rowSum[i] + a[i][j];
                    printf("Sum of row %d = %d\n", i + 1, rowSum[i]);
                }

                printf("The row-sum array contains: [");
                for (i = 0; i < rows; i++)
                {
                    printf("%d", rowSum[i]);
                    if (i < rows - 1) printf(", ");
                }
                printf("]\n");
                break;

            case 3:
                for (i = 0; i < rows; i++)
                    for (j = 0; j < cols; j++)
                        transpose[j][i] = a[i][j];

                printf("Transpose of the matrix:\n");
                for (i = 0; i < cols; i++)
                {
                    for (j = 0; j < rows; j++)
                        printf("%d\t", transpose[i][j]);
                    printf("\n");
                }
                break;

            case 4:
                printf("Exiting program...\n");
                break;

            default:
```

```
        printf("Invalid choice. Please try again.\n");
    }
} while (choice != 4);

return 0;
}
```

4. Execution Screenshot

Terminal execution screenshot provided for the HDOC submission:

```
dhruvpandey@DHRLVs-MacBook-Air ~ % ./day30

===== ARRAY OPERATIONS =====
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Rotate Array Right by k Positions
4. Exit
=====
Enter your choice: 1
Enter number of elements: 2
Enter 2 elements:
3
4
Sorted array: 3 4
Enter element to search: 3
Element 3 found at position 1.

===== ARRAY OPERATIONS =====
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Rotate Array Right by k Positions
4. Exit
=====
Enter your choice: 2
Enter number of elements: 2
Enter 2 elements:
5
6

1. Insertion
2. Deletion
Enter your choice: 1
Enter position for insertion: 1
Enter element to insert: 1
Array after insertion: 1 5 6

===== ARRAY OPERATIONS =====
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Rotate Array Right by k Positions
4. Exit
=====
Enter your choice: 3
Enter number of elements: 2
Enter 2 elements:
3
2
Enter number of positions to rotate: 1
Array after right rotation: 2 3

===== ARRAY OPERATIONS =====
1. Bubble Sort and Binary Search
2. Insertion and Deletion
3. Rotate Array Right by k Positions
4. Exit
=====
Enter your choice: █
```

5. Result

The menu-driven C program for 2D array operations was prepared successfully. It calculates the sum of all matrix elements, stores and displays the sum of each row, finds the transpose of the matrix, and provides an exit option.